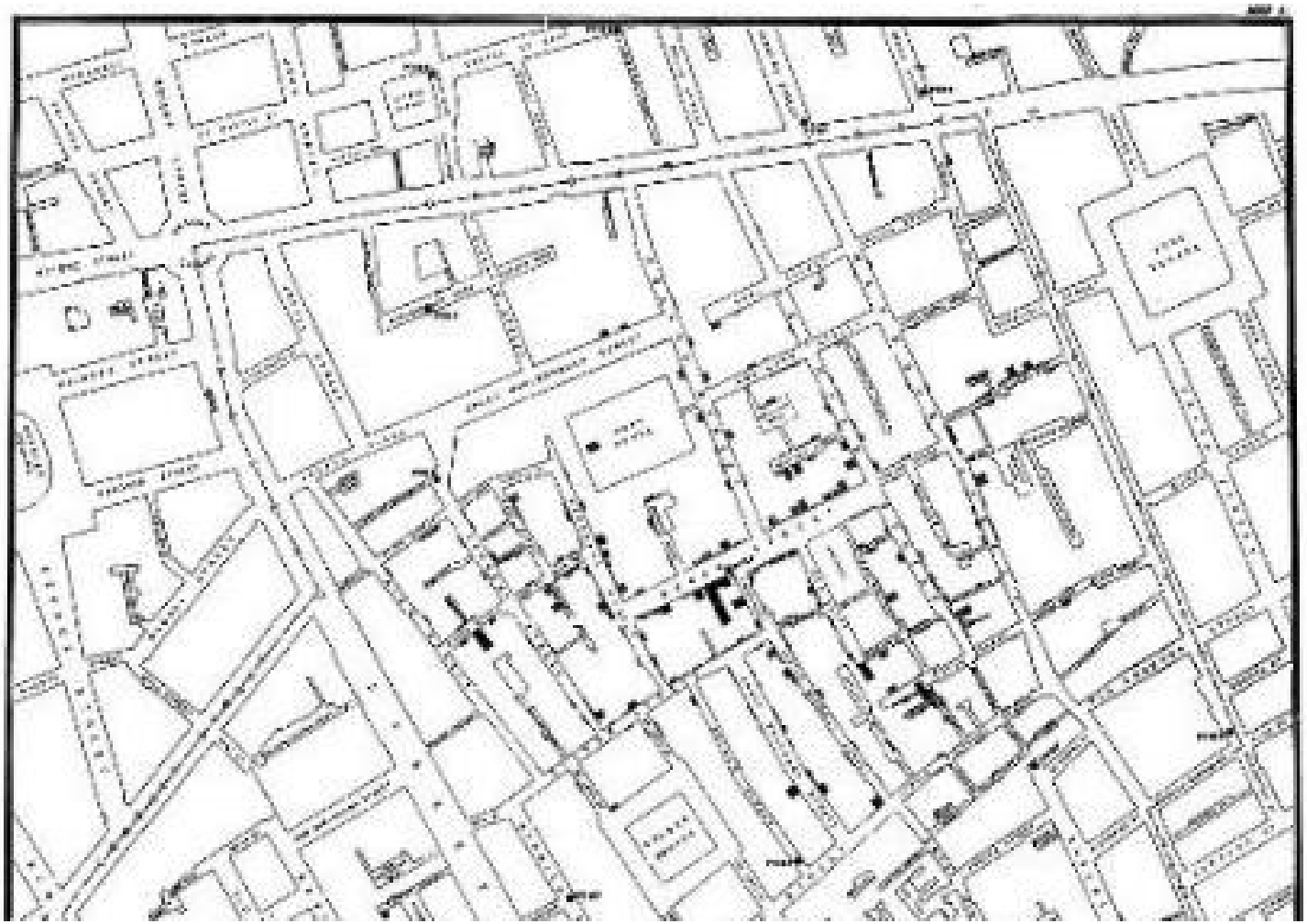
What is Geographic Information Systems (GIS)?



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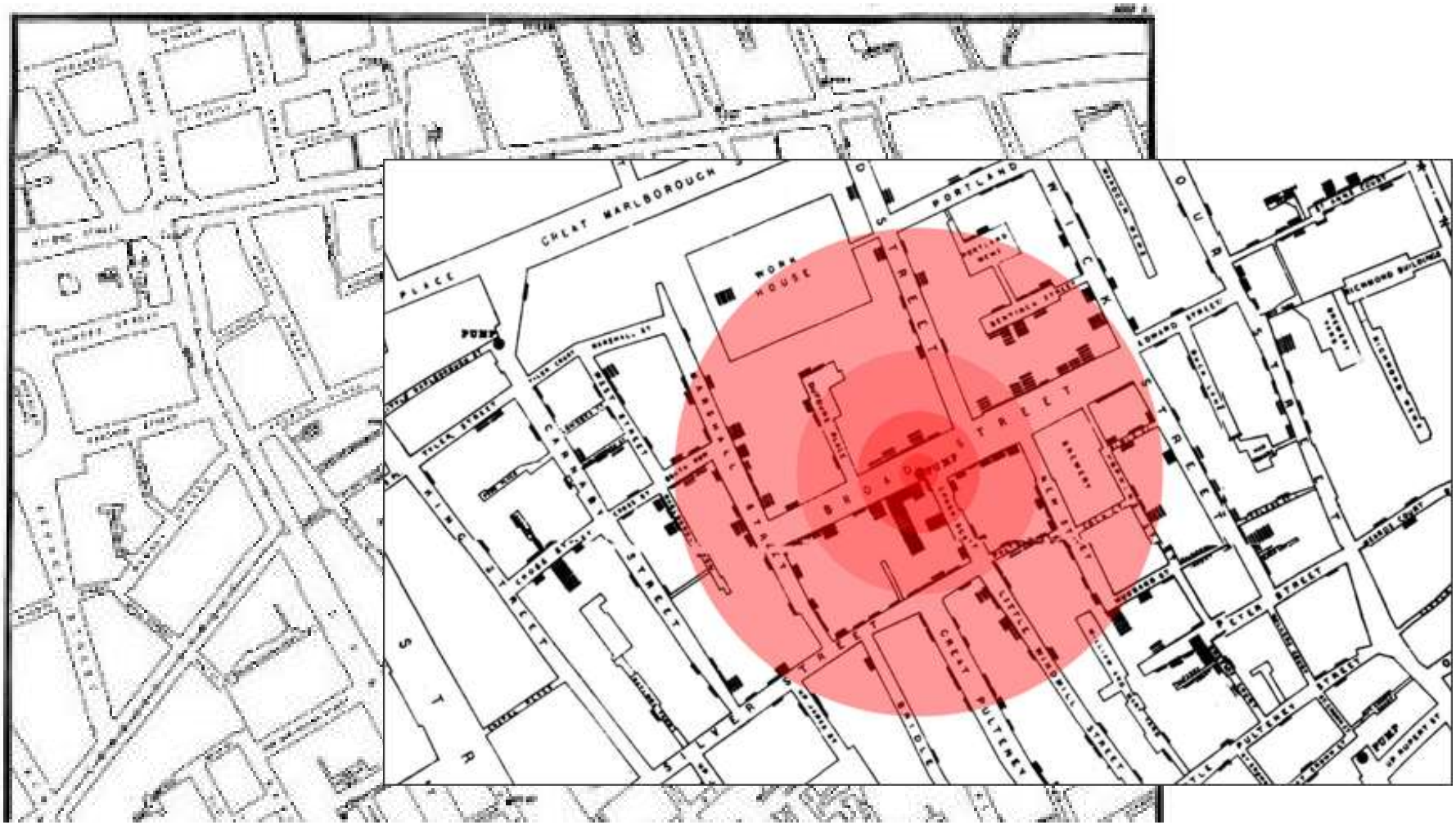
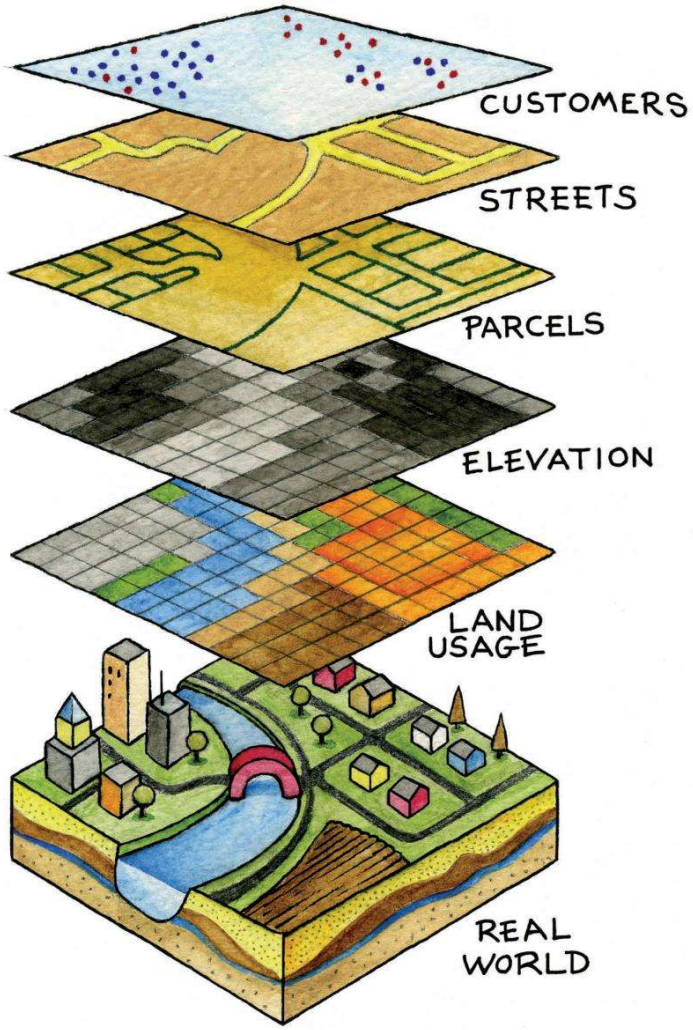


What is Geographic Information Systems (GIS)

Unique content management system that can store data by attaching layers of descriptive information to absolute location as well as to a database

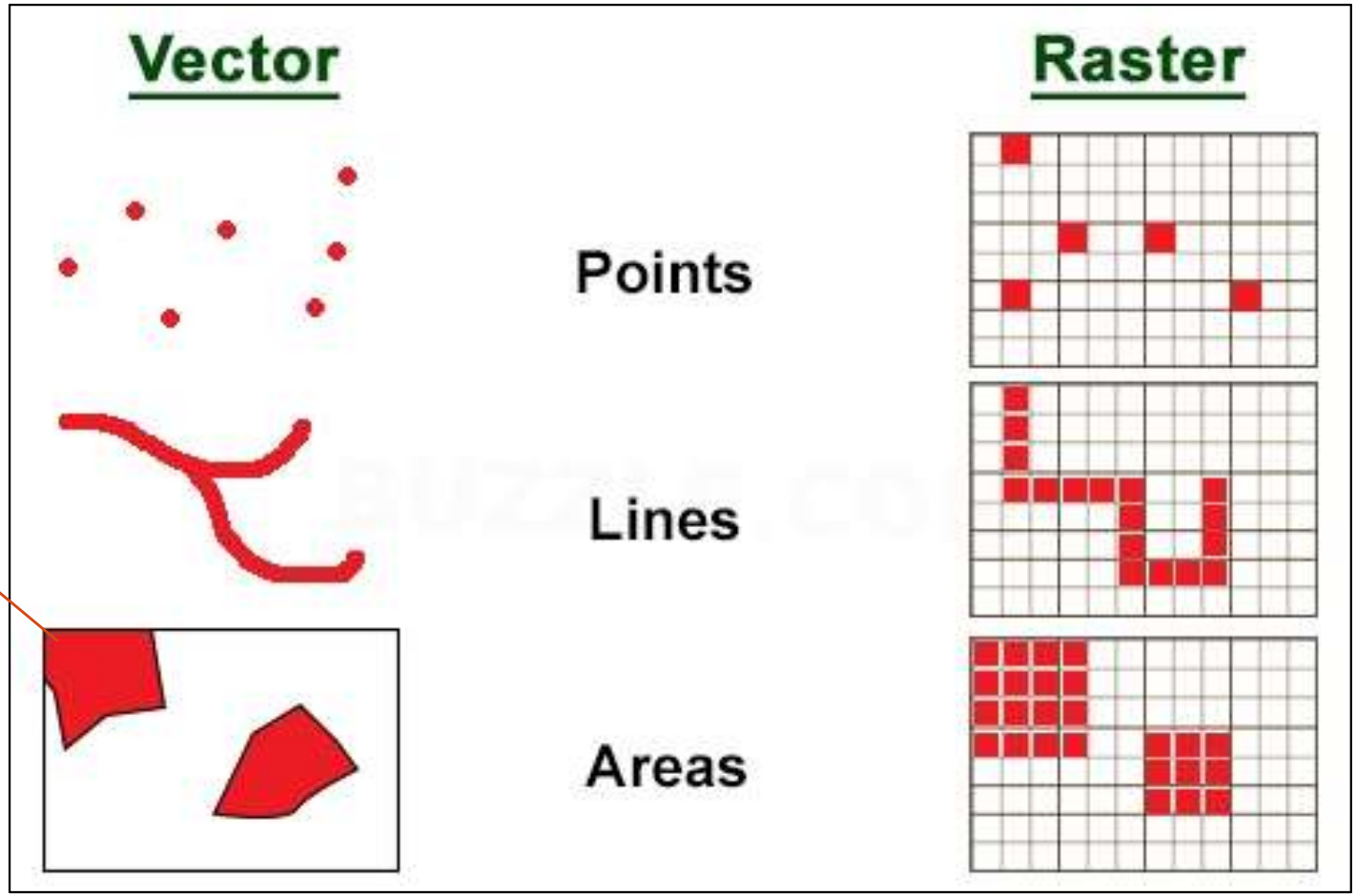
Among its unique benefits is the ability to relate otherwise disparate data sources on the basis of common geography, revealing patterns that are otherwise invisible



GIS Data Types

Tabular

shape	quantity
1	30
2	50
3	1,00



So many technologies

Processing Tools

- **ArcGIS**
- Quantum GIS (**QGIS**)
- Open GeoDa
- IDRISI
- ERDAS Imagine
- R
- Microsoft Excel (yes)

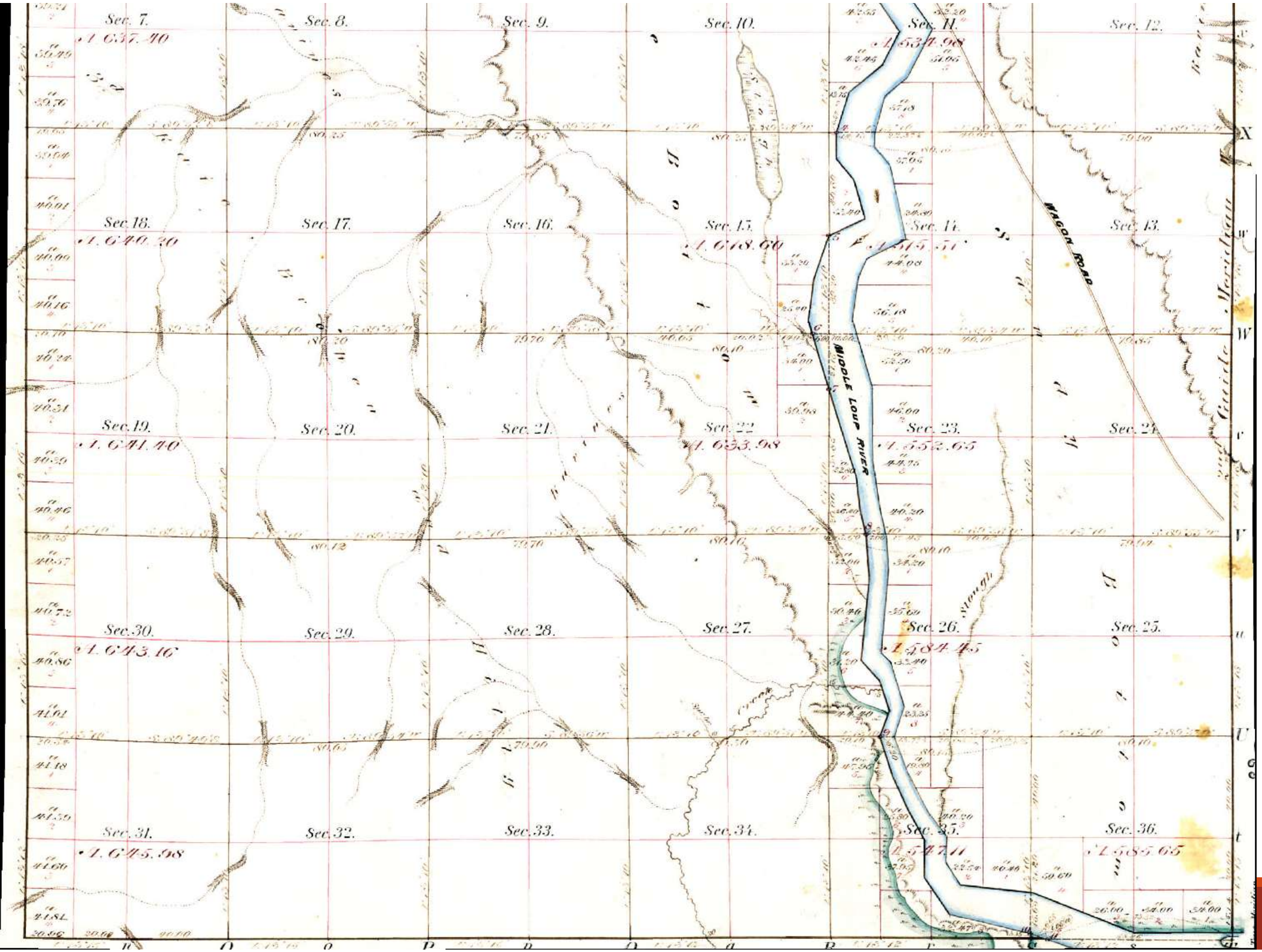
And many GIS file types

Vector – *selected examples*

- Esri Shapefile (.shp, .dbf, .shx, etc)
- GeoJSON
- Geography Markup Language (GML)
- Keyhole Markup Language (KML/KMZ)
- GPS eXchange Format (.GPX)
- *Tabular data (e.g. CSV) can be turned into vector data formats if they contain columns with latitude and longitude*

Raster – *selected examples*

- GeoTIFF (.TIF, .TIFF)
- ASCII Grid (.ASC)
- Esri Grid
- MrSID
- Envi Raster (.BIL, .BIP, .BSQ)
- Idrisi Raster (.RST, .RDC)
- *Image files (such as .JP2, .JPEG, etc) with an accompanying "world file" (e.g. .TFW)*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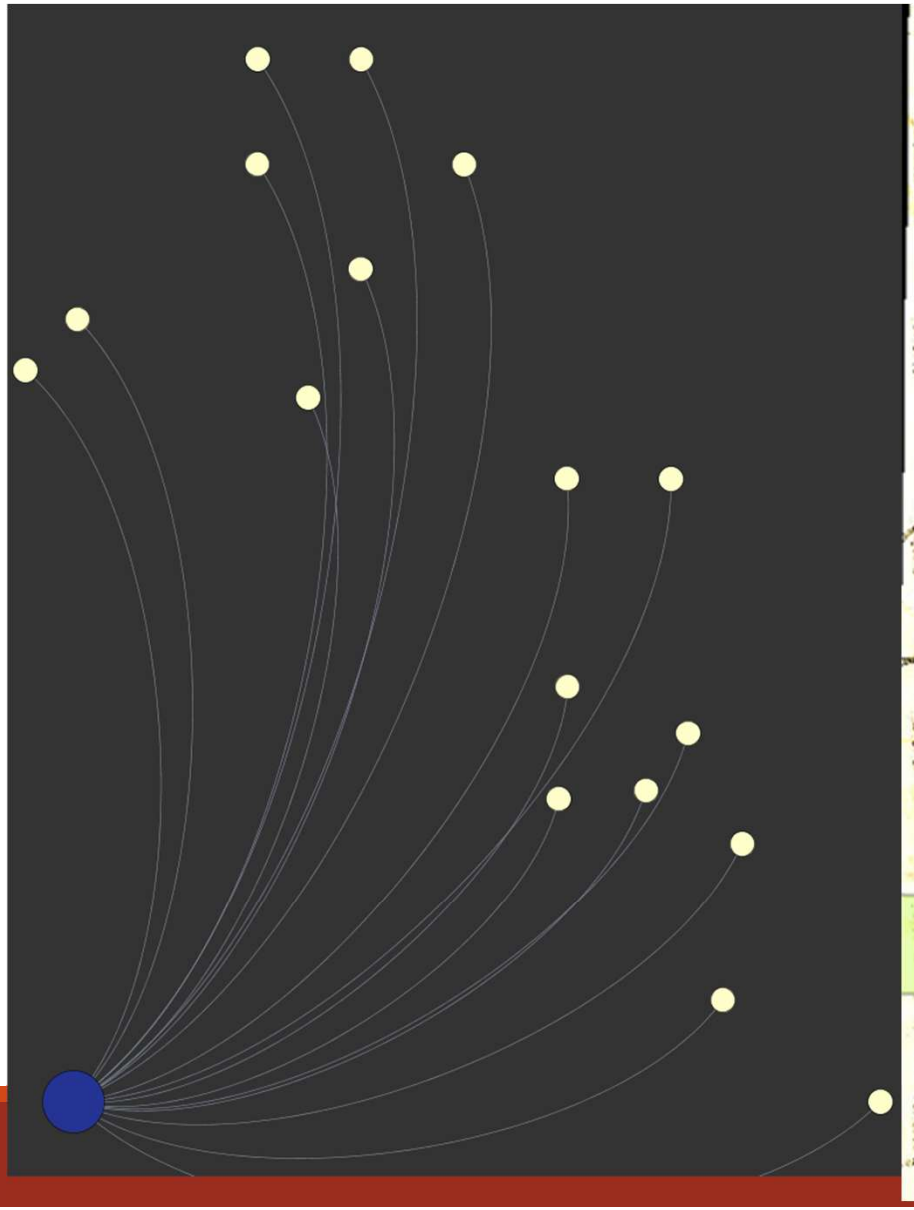
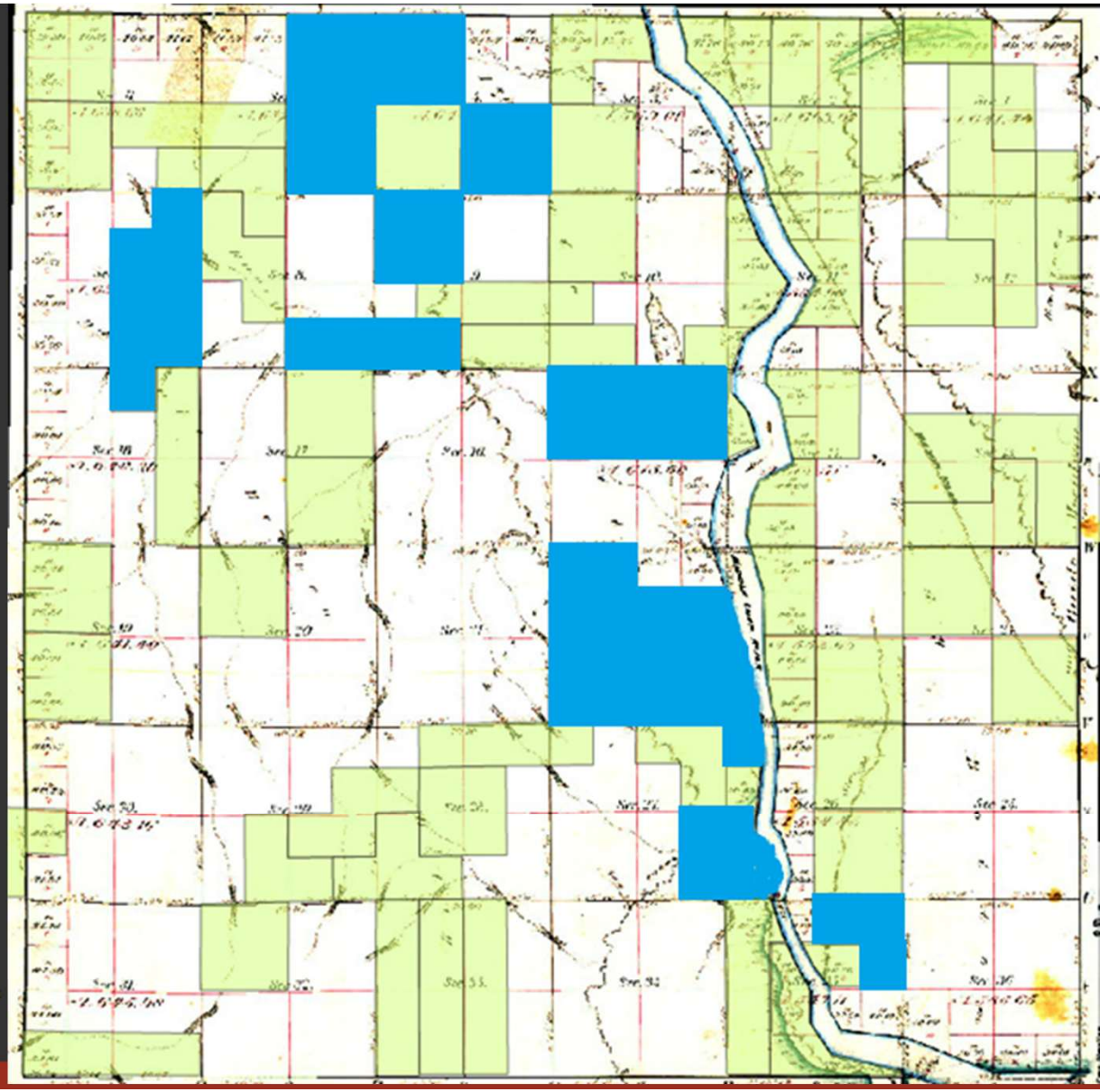
Sec. 26.
Sec. 25.
Sec. 31.
Sec. 32.
Sec. 33.
Sec. 34.
Sec. 35.
Sec. 36.
MIDDLE LOUP RIVER
WAGON ROAD
Slough
Creek

Geolocated social networks



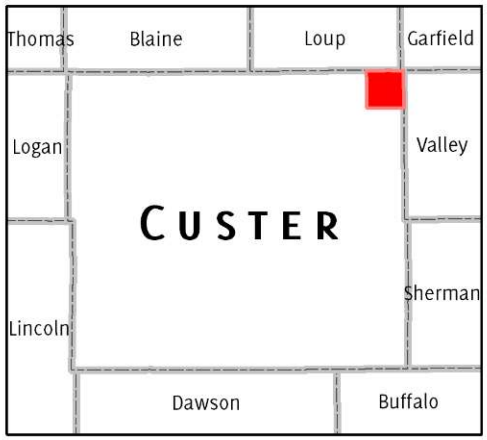


APPLICANT

Place of Origin

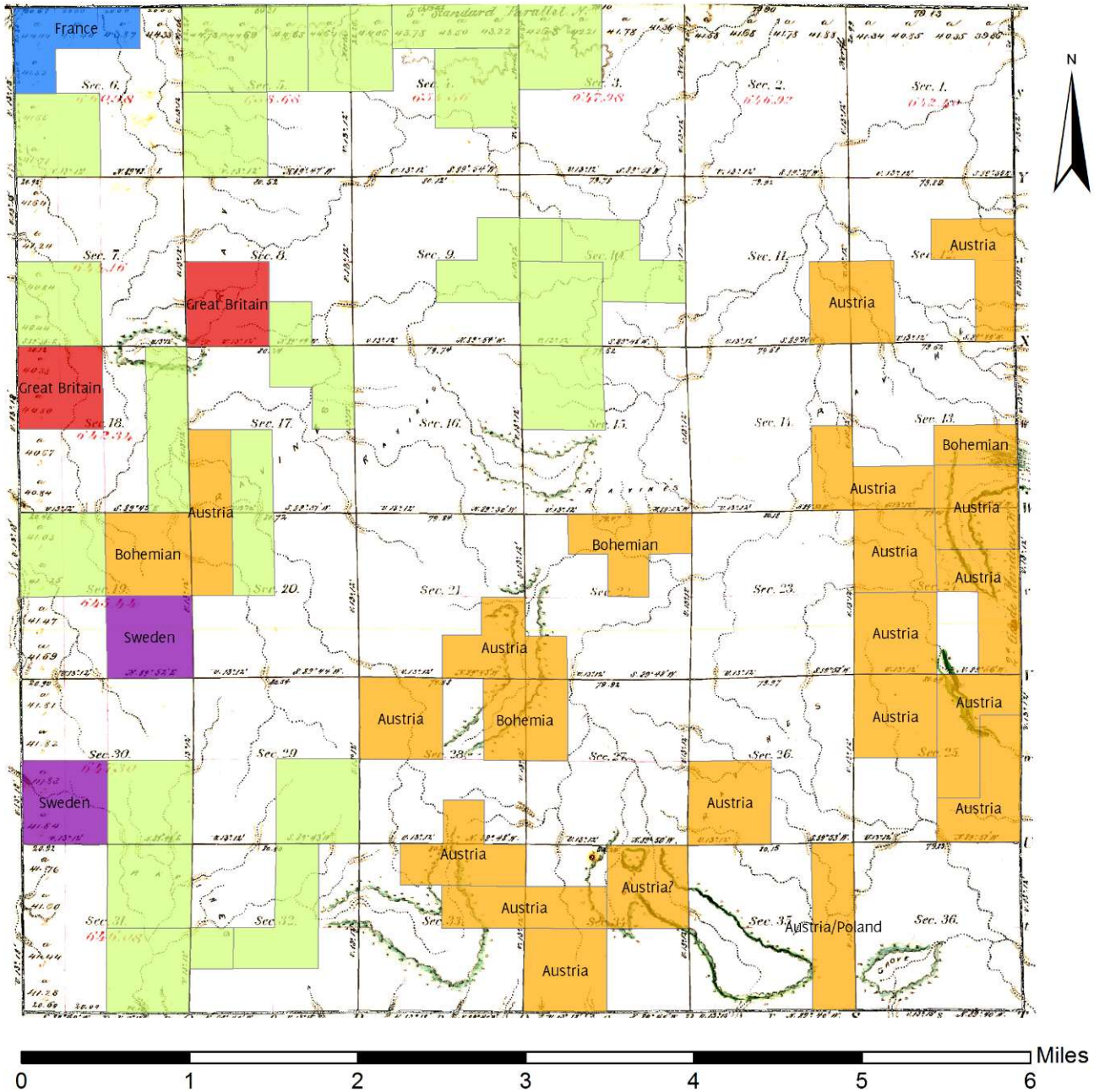
- **Central Europe** *Austria, Bohemia, Moravia*
- **United States**
- **United Kingdom**
- **France**
- **Sweden**

Location within Custer County



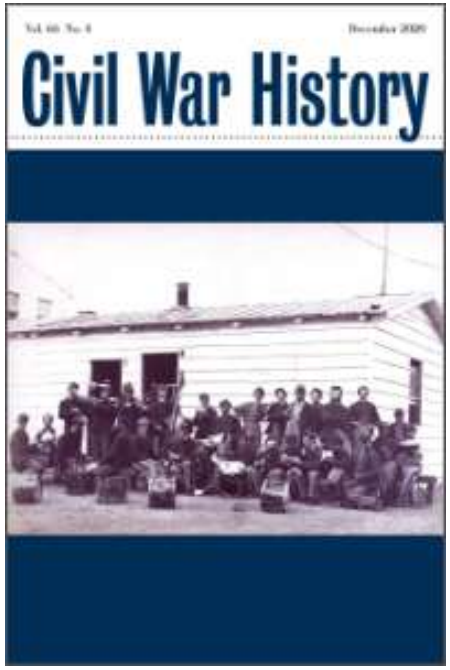
Base map is the original land survey, georeferenced by map author.

State Plane Nebraska (2600) coordinate system, using Lambert Conformal Conic Projection



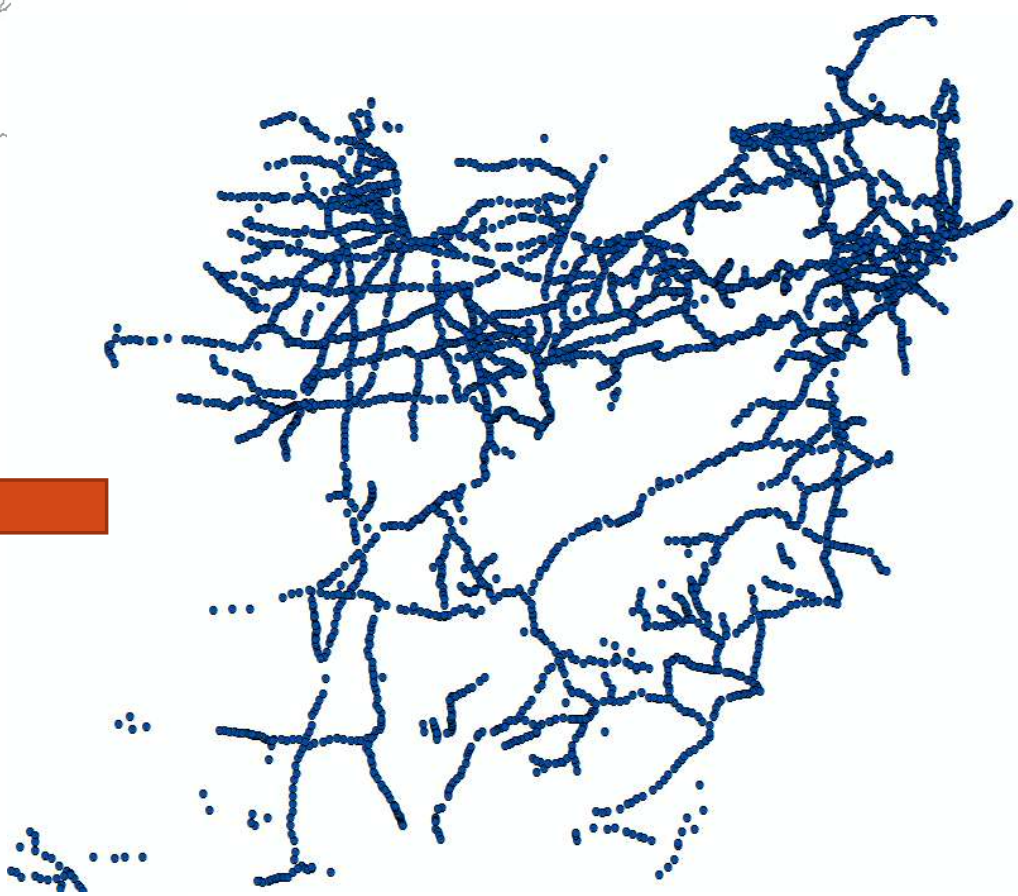
Center for Great Plains Studies | Center for Digital Research in the Humanities | Rob Shepard, Cartographer

Railroads and the Making of Modern America +



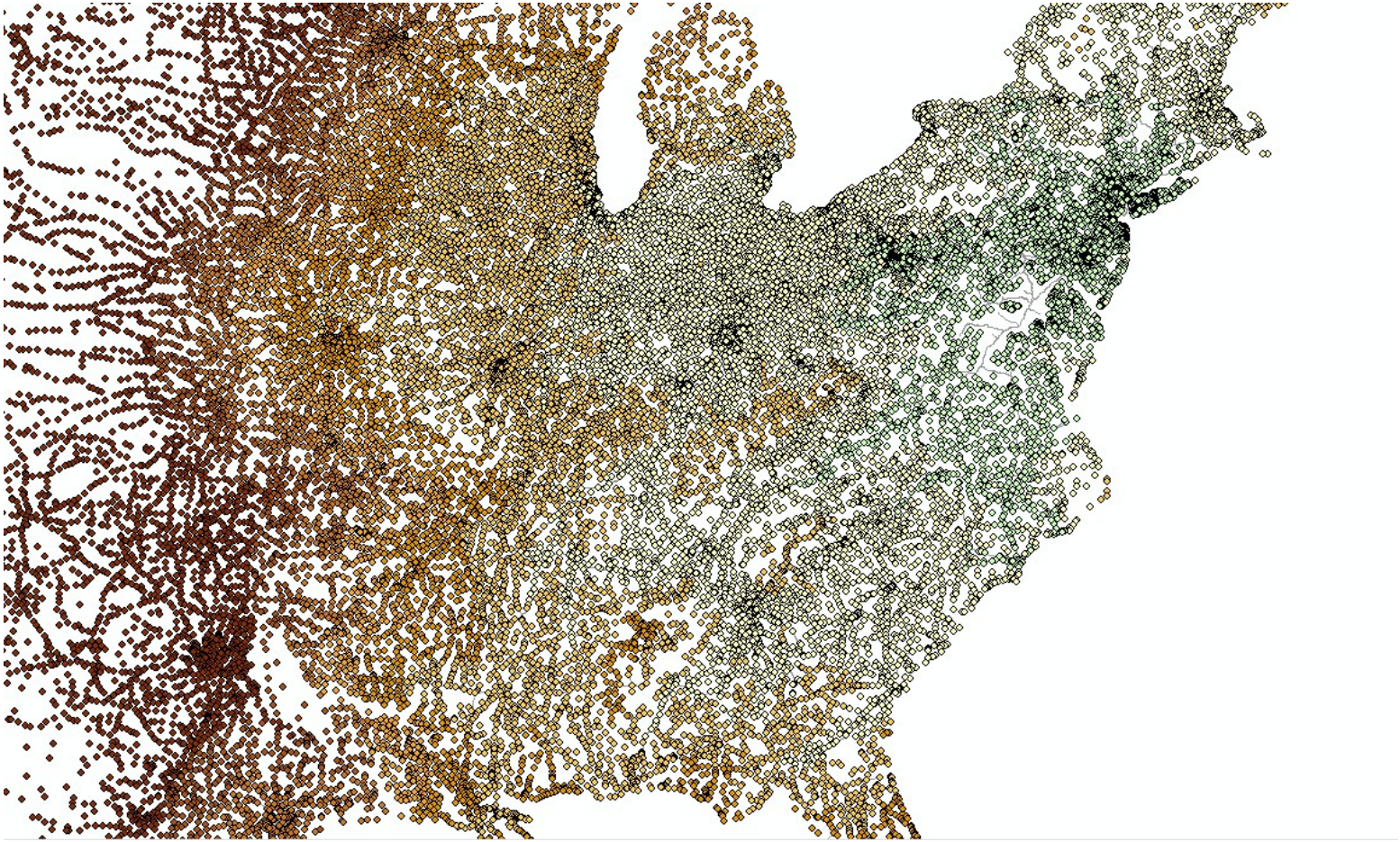
GIS PROCESSING: AN EXAMPLE FROM ARCGIS

Data – Railroads and the Making of Modern America

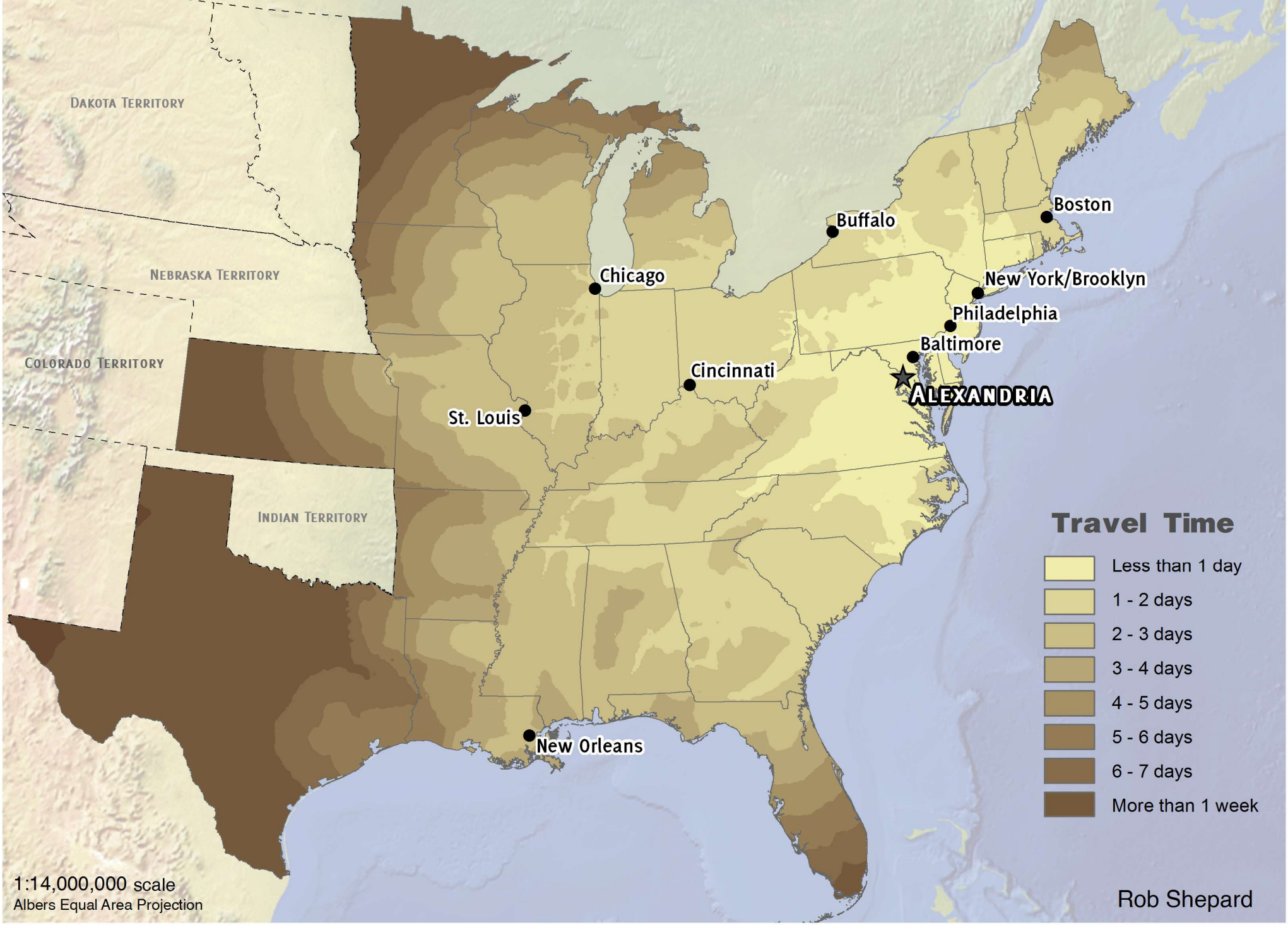


Network analysis: trace/geometric network

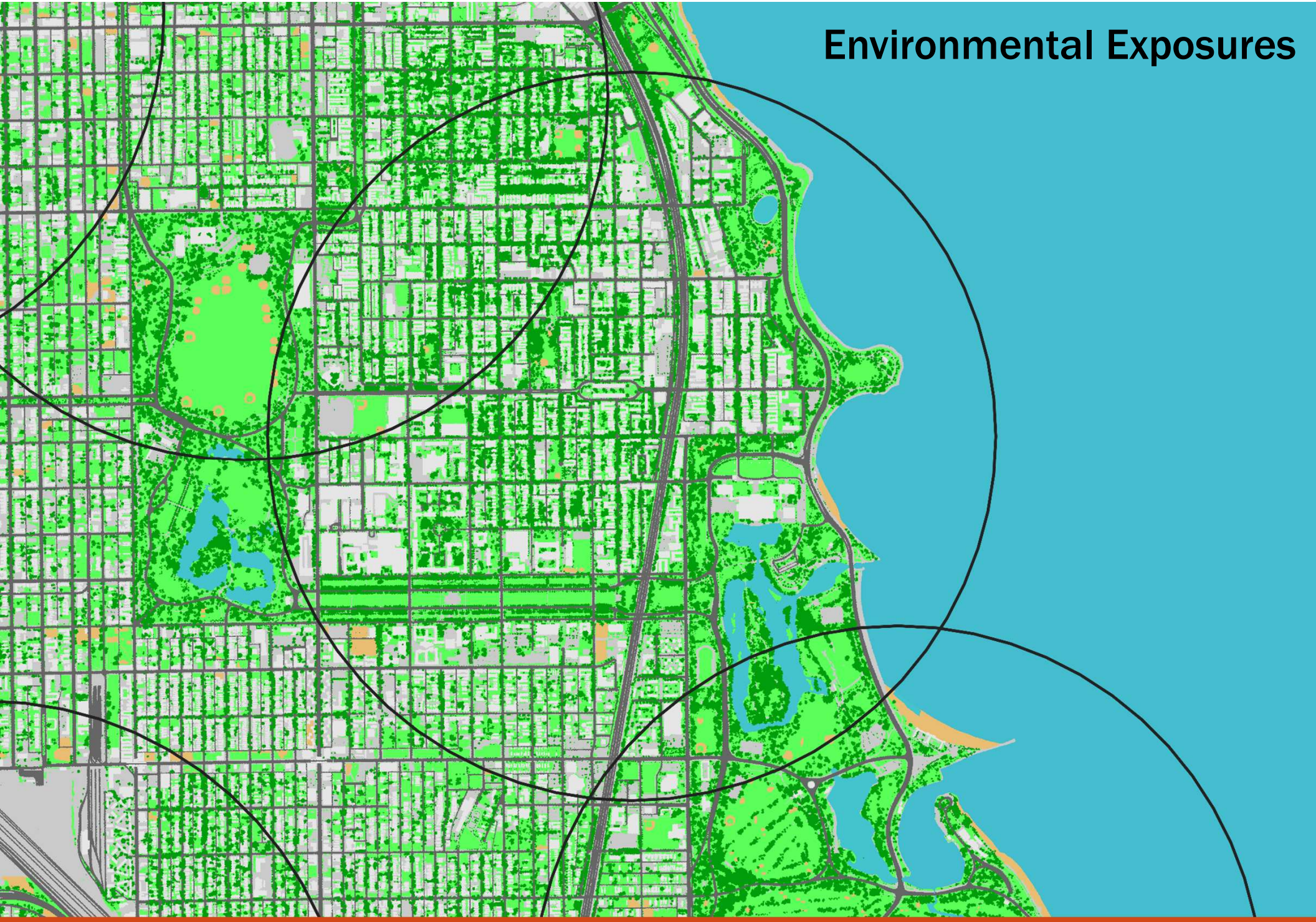
data generation



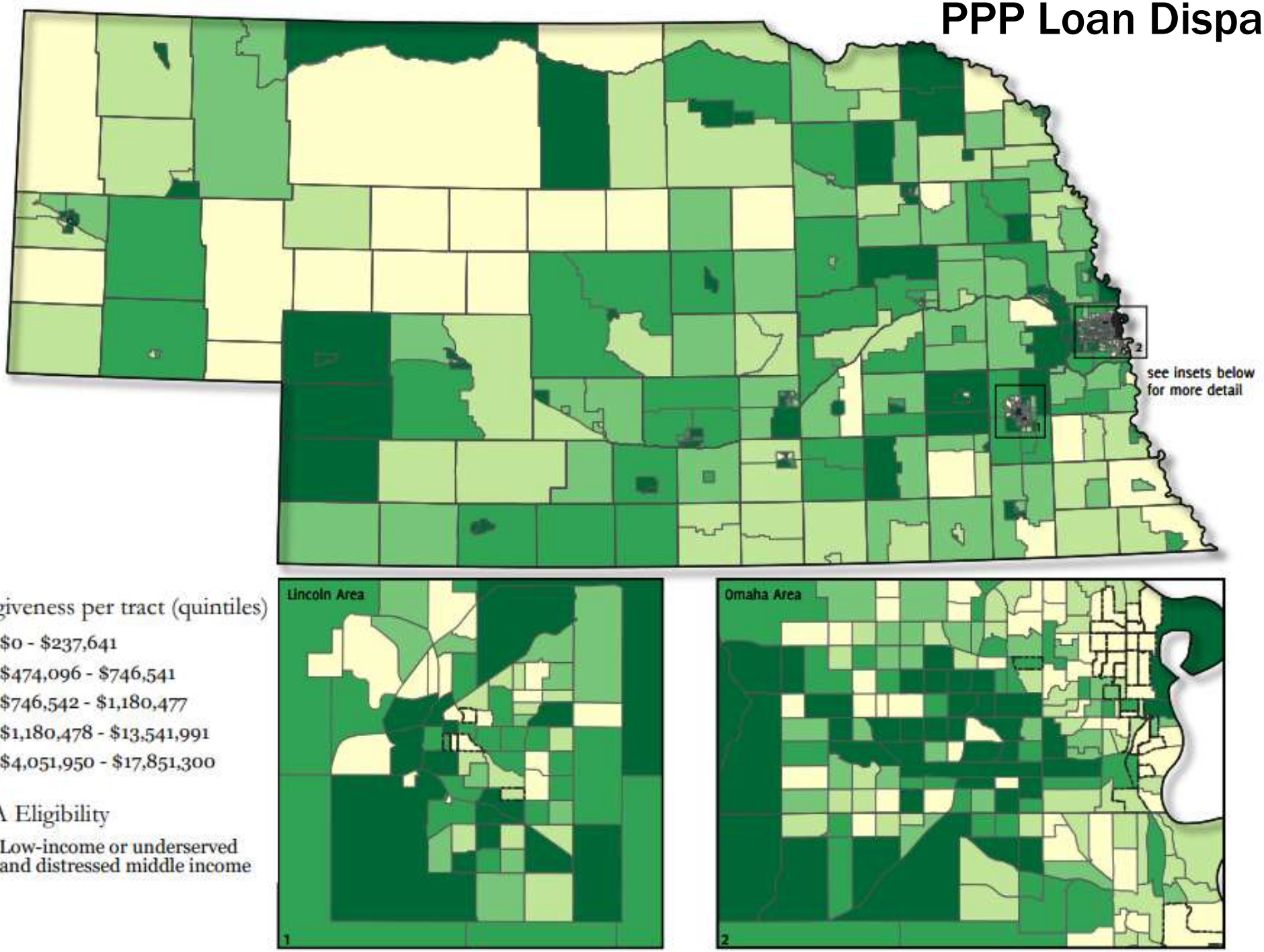
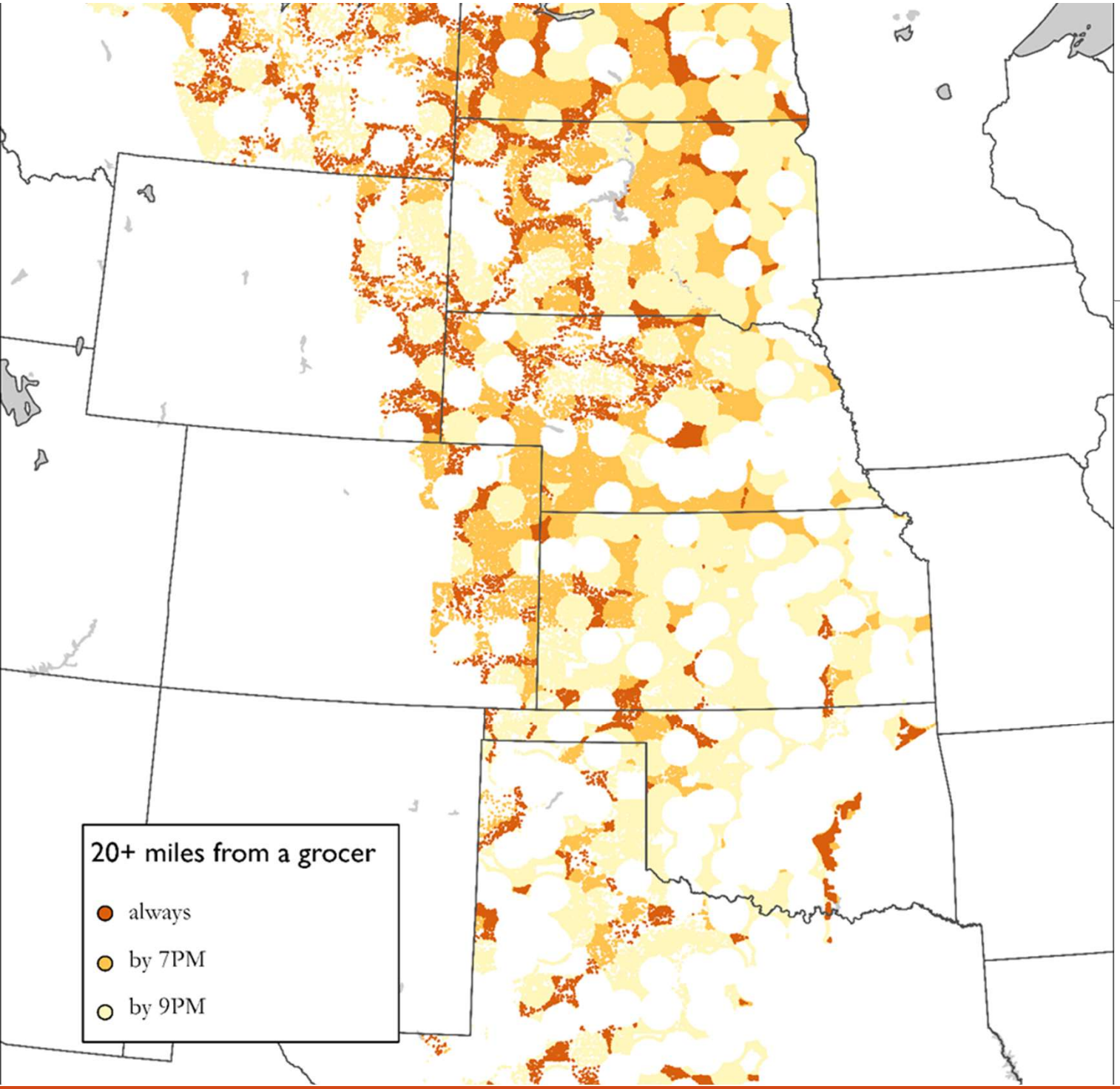
Railroad Rates of Travel from Alexandria, 1861



Environmental Exposures



PPP Loan Disparities



Grocery store access